

MULTIPLE CHOICE QUESTIONS (MCQs)

Chapter: Queue and Deque (Data Structures)

Introduction to Queue

1. Which data structure was studied in the previous chapter before Queue?
 - A. Array
 - B. List
 - C. Stack
 - D. Tree
2. Stack follows which principle?
 - A. FIFO
 - B. LILO
 - C. LIFO
 - D. FCFS
3. Queue follows which principle?
 - A. Last In First Out
 - B. First In Last Out
 - C. First In First Out
 - D. Random In Random Out
4. Queue is best described as:
 - A. An unordered collection
 - B. An ordered linear list
 - C. A hierarchical structure
 - D. A non-linear structure
5. In a queue, insertion and deletion take place:
 - A. From the same end
 - B. From any end
 - C. From different ends
 - D. From the middle
6. Which of the following is a real-life example of a queue?
 - A. Stack of books
 - B. Bank cash counter line
 - C. Tree branches
 - D. Graph traversal
7. Vehicles waiting at a fuel pump demonstrate which data structure?
 - A. Stack
 - B. Tree
 - C. Queue
 - D. Graph

FIFO Principle

8. FIFO stands for:
 - A. First In First Out
 - B. Fast In Fast Out
 - C. First In Free Out
 - D. Fixed In Fixed Out
 9. FIFO is also known as:
 - A. Last Come First Served
 - B. First Come First Served
 - C. Priority Scheduling
 - D. Round Robin
 10. The element removed first from a queue is:
 - A. The most recent element
 - B. The middle element
 - C. The element inserted last
 - D. The element that has stayed longest
 11. New elements in a queue are added at:
 - A. Front
 - B. Middle
 - C. Rear
 - D. Any position
 12. Elements are removed from a queue at:
 - A. Rear
 - B. Middle
 - C. Front
 - D. Random position
 13. REAR of a queue is also known as:
 - A. HEAD
 - B. FRONT
 - C. TAIL
 - D. ROOT
 14. FRONT of a queue is also called:
 - A. TAIL
 - B. HEAD
 - C. REAR
 - D. LEAF
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Applications of Queue (Real Life)

15. Train tickets in waiting list are managed using:
 - A. Stack
 - B. Queue
 - C. Deque
 - D. Tree

16. W/L1 ticket gets confirmed when:
- A. Any ticket is cancelled
 - B. The last ticket is cancelled
 - C. A confirmed ticket is cancelled
 - D. New ticket is booked
17. IVRS customer support system uses which structure?
- A. Stack
 - B. Tree
 - C. Queue
 - D. Graph
18. Single-lane one-way traffic follows:
- A. LIFO
 - B. FIFO
 - C. Random order
 - D. Priority order
19. Vehicles at a toll booth are served based on:
- A. Priority
 - B. Stack
 - C. FIFO
 - D. Recursion
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Applications of Queue in Computer Science

20. A web server handling limited concurrent requests uses:
- A. Stack
 - B. Queue
 - C. Deque
 - D. Graph
21. In multitasking operating systems, jobs are:
- A. Executed randomly
 - B. Executed simultaneously
 - C. Queued for CPU
 - D. Stored permanently
22. Processor can handle how many tasks at a time?
- A. Zero
 - B. One
 - C. Two
 - D. Unlimited
23. Print requests from multiple computers are handled using:
- A. Stack
 - B. Queue
 - C. Tree
 - D. Heap

24. FIFO scheduling is the simplest scheduling because:
- A. It uses recursion
 - B. It processes tasks randomly
 - C. It follows order of arrival
 - D. It uses priority
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Operations on Queue

25. Which operation inserts an element into a queue?
- A. PUSH
 - B. POP
 - C. ENQUEUE
 - D. INSERTFRONT
26. ENQUEUE operation inserts element at:
- A. Front
 - B. Rear
 - C. Middle
 - D. Any position
27. Which operation removes an element from queue?
- A. PUSH
 - B. POP
 - C. DEQUEUE
 - D. INSERT
28. DEQUEUE removes element from:
- A. Rear
 - B. Front
 - C. Middle
 - D. End
29. Overflow occurs when:
- A. Queue is empty
 - B. Queue is full
 - C. Queue has one element
 - D. Queue is dynamic
30. Underflow occurs when:
- A. Inserting in full queue
 - B. Deleting from empty queue
 - C. Reading from full queue
 - D. Inserting in empty queue
31. Which operation checks whether queue is empty?
- A. IS FULL
 - B. IS EMPTY
 - C. PEEK
 - D. SIZE

32. PEEK operation is used to:

- A. Insert element
- B. Delete element
- C. View front element
- D. View rear element

33. IS FULL operation avoids:

- A. Underflow
- B. Overflow
- C. Deletion
- D. Traversal

Queue Representation and Behavior

34. In a simple queue diagram, Front is shown on:

- A. Right side
- B. Left side
- C. Middle
- D. Bottom

35. Rear of queue is shown on:

- A. Left side
- B. Right side
- C. Top
- D. Bottom

36. ENQUEUE always occurs at:

- A. Front
- B. Rear
- C. Middle
- D. Both ends

Implementation of Queue using Python

37. Which Python data type is used to implement queue?

- A. Tuple
- B. Dictionary
- C. List
- D. Set

38. Queue is created using:

- A. myQueue = {}
- B. myQueue = ()
- C. myQueue = list()
- D. myQueue = set()

39. Which method is used to add element at rear?

- A. insert()
- B. append()

- C. pop()
- D. remove()

40. append() adds element:

- A. At index 0
- B. At middle
- C. At end of list
- D. Randomly

41. Python does not require IS FULL check because:

- A. Python is compiled
- B. Python lists are dynamic
- C. Python uses arrays
- D. Python has fixed size lists

42. isEmpty() function checks:

- A. Queue capacity
- B. Queue elements
- C. Queue memory
- D. Queue index

43. isEmpty() returns True when:

- A. Queue has elements
- B. Queue length is zero
- C. Queue is full
- D. Queue is dynamic

44. Which method deletes element from front?

- A. pop()
- B. pop(0)
- C. remove()
- D. delete()

45. pop(0) removes element from:

- A. End of list
- B. Middle of list
- C. Beginning of list
- D. Random index

46. size of queue is found using:

- A. count()
- B. length()
- C. size()
- D. len()

47. peek() returns:

- A. Last element
- B. Middle element
- C. First element
- D. Random element

48. peek() returns None when:
- A. Queue is full
 - B. Queue is empty
 - C. Queue has one element
 - D. Queue is dynamic
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Queue Example Scenario (Bank)

49. Two friends joining queue performs:
- A. dequeue twice
 - B. enqueue twice
 - C. peek twice
 - D. size twice
50. Cashier calling “Next” implies:
- A. enqueue
 - B. peek
 - C. dequeue
 - D. size
51. Checking number of people in queue uses:
- A. peek()
 - B. isEmpty()
 - C. size()
 - D. enqueue()
52. Removing all people until empty leads to:
- A. Overflow
 - B. Normal exit
 - C. Underflow
 - D. Deadlock
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Introduction to Deque

53. Deque stands for:
- A. Direct queue
 - B. Double ended queue
 - C. Dynamic queue
 - D. Distributed queue
54. Deque allows insertion and deletion:
- A. From front only
 - B. From rear only
 - C. From both ends
 - D. From middle
55. Deque can behave as:
- A. Stack only
 - B. Queue only

- C. Both stack and queue
- D. Tree

56. Deque is pronounced as:
- A. Duke
 - B. Deck
 - C. Dee-queue
 - D. Duck

Applications of Deque

57. Rejoining queue from front demonstrates:
- A. Stack
 - B. Queue
 - C. Deque
 - D. Tree
58. Vehicles shifting between toll booths use:
- A. Stack
 - B. Queue
 - C. Deque
 - D. Graph
59. Browser history uses deque because:
- A. URLs are random
 - B. Oldest URLs are removed
 - C. New URLs are not stored
 - D. History is infinite
60. Undo/Redo feature uses:
- A. Stack
 - B. Queue
 - C. Deque
 - D. Tree
61. Palindrome checking uses deque because:
- A. Characters are compared randomly
 - B. Characters are accessed from both ends
 - C. Deque sorts characters
 - D. Deque reverses string

Operations on Deque

62. INSERTFRONT inserts element at:
- A. Rear
 - B. Middle
 - C. Front
 - D. End

63. INSERTREAR operation is similar to:
- A. Stack push
 - B. Queue enqueue
 - C. Queue dequeue
 - D. Stack pop
64. DELETIONFRONT removes element from:
- A. Rear
 - B. Middle
 - C. Front
 - D. Any end
65. DELETIONREAR removes element from:
- A. Front
 - B. Middle
 - C. Rear
 - D. Any end
66. If insertion and deletion occur at same end, deque behaves as:
- A. Queue
 - B. Stack
 - C. List
 - D. Tree
67. If insertion and deletion occur at opposite ends, deque behaves as:
- A. Stack
 - B. Queue
 - C. Tree
 - D. Graph

Palindrome Algorithm using Deque

68. Characters are inserted using:
- A. INSERTFRONT
 - B. INSERTREAR
 - C. DELETIONFRONT
 - D. PEEK
69. Palindrome checking removes characters from:
- A. Front only
 - B. Rear only
 - C. Both ends
 - D. Middle
70. String is palindrome if:
- A. Characters differ
 - B. Deque has many elements
 - C. Characters match till empty or one left
 - D. Queue is full

Implementation of Deque using Python

71. Deque is implemented using:

- A. Tuple
- B. Set
- C. List
- D. Dictionary